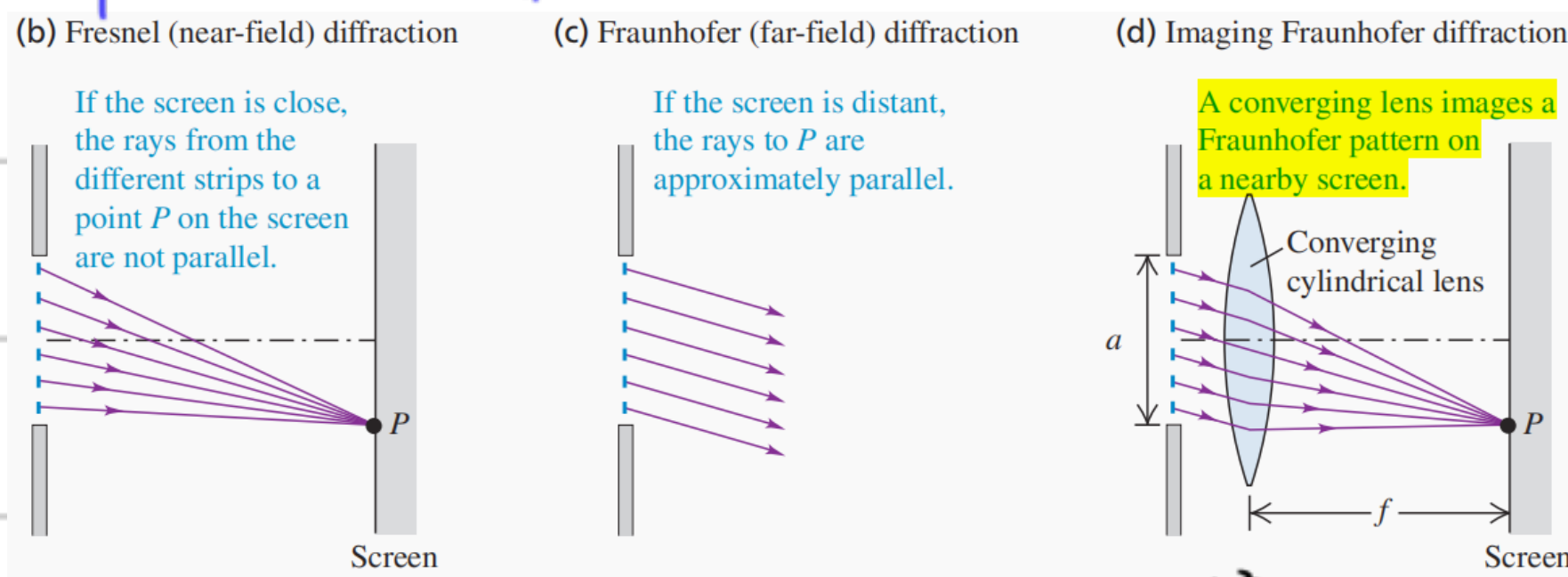
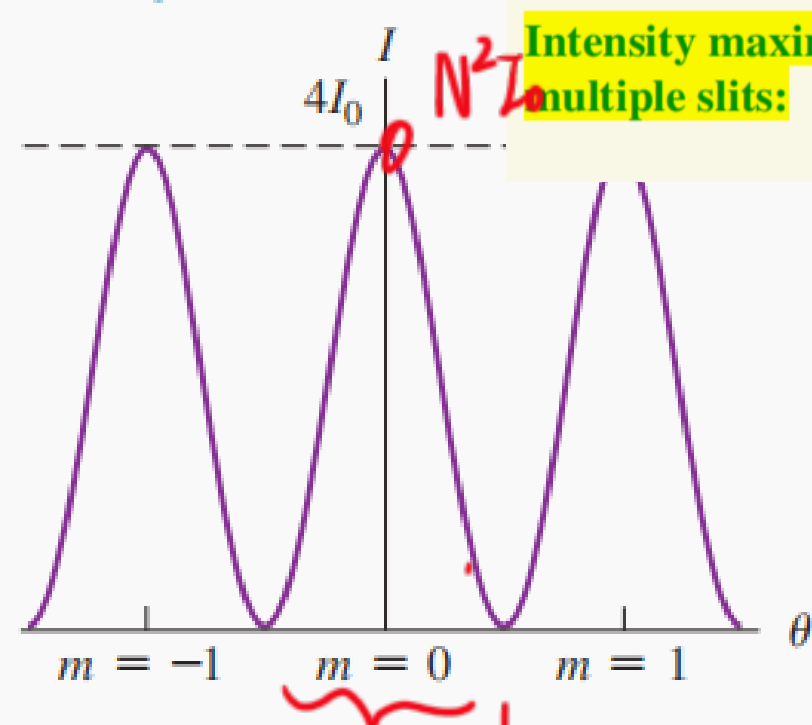


Chap 36 Diffraction



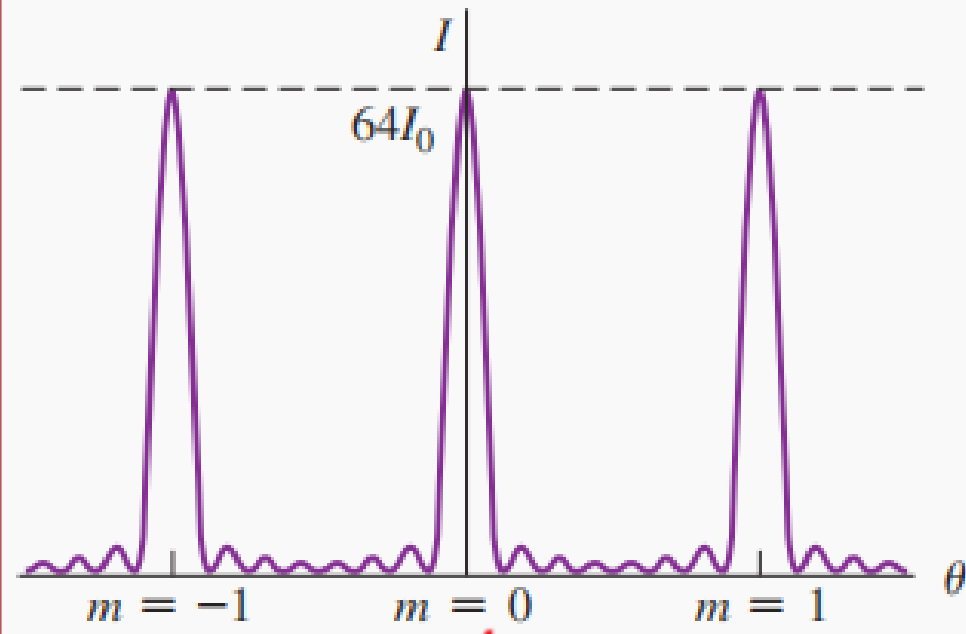
(a) $N = 2$: two slits produce one minimum between adjacent maxima.



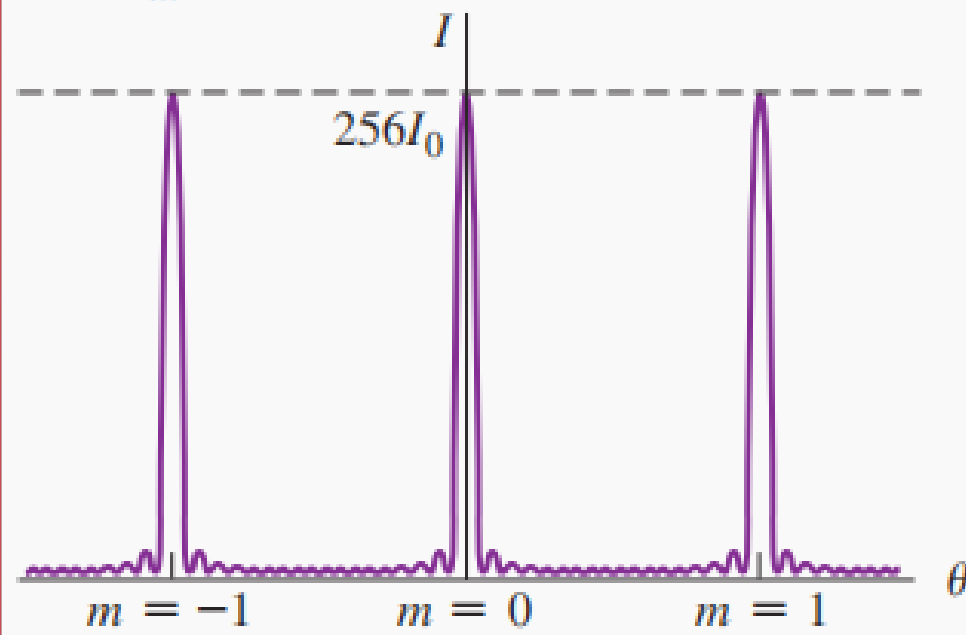
Distance between slits d Wavelength λ

多缝衍射的亮条纹

(b) $N = 8$: eight slits produce taller, narrower maxima in the same locations, separated by seven minima.



(c) $N = 16$: with 16 slits, the maxima are even taller and narrower, with more intervening minima.



You should show that when there are N slits, there are $(N - 1)$ minima between each pair of principal maxima and a minimum occurs whenever ϕ is an integer multiple of $2\pi/N$ (except when ϕ is an integer multiple of 2π , which gives a principal maximum).

chromatic resolving power R (不是 resolving power)

$$R = \frac{\lambda}{\Delta\lambda} = Nm \text{ (chromatic resolving power)}$$

$\Rightarrow$ how well different λ can be distinguished by diffraction grating

Angular radius of first dark ring = angular radius of Airy disk

$$\sin \theta_1 = 1.22 \frac{\lambda}{D}$$

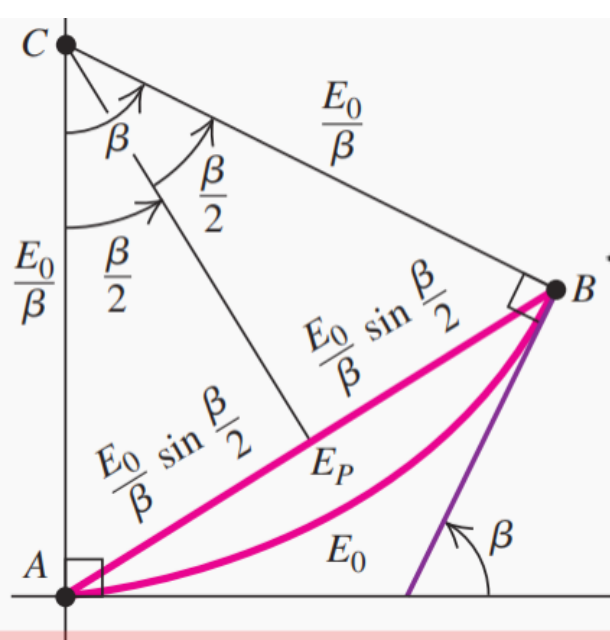
Wavelength λ Aperture diameter D

联系 Rayleigh's criterion
 $\lambda \downarrow \theta \downarrow$ DVD 上 slits $\uparrow$ information $\uparrow$
 $\rightarrow$ 圆孔衍射暗条纹

slit 越窄, diffraction pattern 越宽
 单缝衍射 暗条纹

Dark fringes, single-slit diffraction:
 Angle of line from center of slit to m th dark fringe on screen
 $\sin \theta = \frac{m\lambda}{a}$ ($m = \pm 1, \pm 2, \pm 3, \dots$)
 Slit width a Wavelength λ

$$y_m = \lambda \frac{m\lambda}{a} \text{ (对 } y_m \ll \lambda \text{)}$$



$\rightarrow$ phasor diagram 推导出



$$E_P = E_0 \frac{\sin(\beta/2)}{\beta/2} \text{ (amplitude in single-slit diffraction)} \quad (36.4)$$

The intensity at each point on the screen is proportional to the square of the amplitude given by Eq. (36.4). If I_0 is the intensity in the straight-ahead direction where $\theta = 0$ and $\beta = 0$, then the intensity I at any point is

$$I = I_0 \left[\frac{\sin(\beta/2)}{\beta/2} \right]^2 \text{ (intensity in single-slit diffraction)} \quad (36.5)$$

Angle of line from center of slit to position on screen

Intensity in single-slit diffraction

$$I = I_0 \left\{ \frac{\sin [\pi a (\sin \theta) / \lambda]}{\pi a (\sin \theta) / \lambda} \right\}^2$$

Intensity at $\theta = 0$ Slit width a Wavelength λ

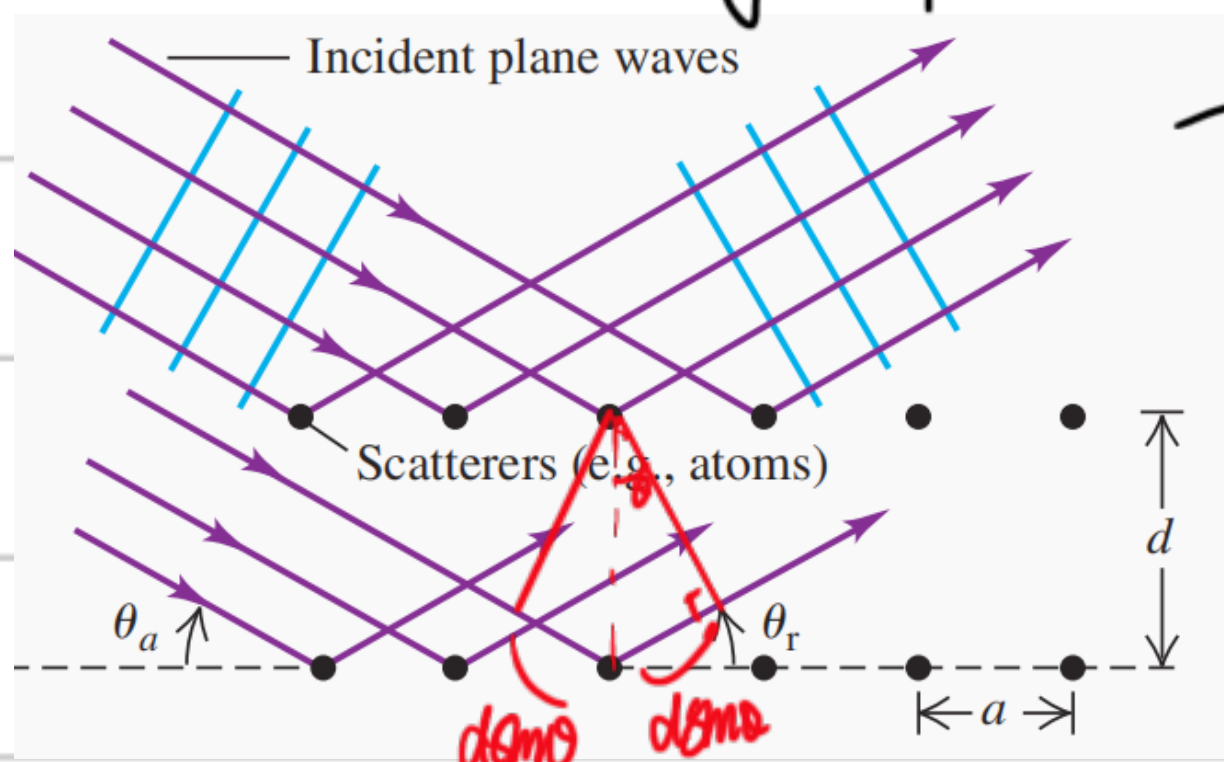
解释声波穿透和光波不穿透

Figure 36.11 The sound waves used in speech have a long wavelength (about 1 m) and can easily bend around this instructor's head. By contrast, light waves have very short wavelengths and undergo very little diffraction. Hence you can't see around his head!

Diffraction by a circular aperture:

Figure 36.17 Microscopic pits on the surface of this DVD act as a reflection grating, splitting white light into its component colors.

X-Ray Bragg reflection is truly interference



Bragg condition for constructive interference from an array:

$$2d \sin \theta = m\lambda \quad (m = 1, 2, 3, \dots)$$

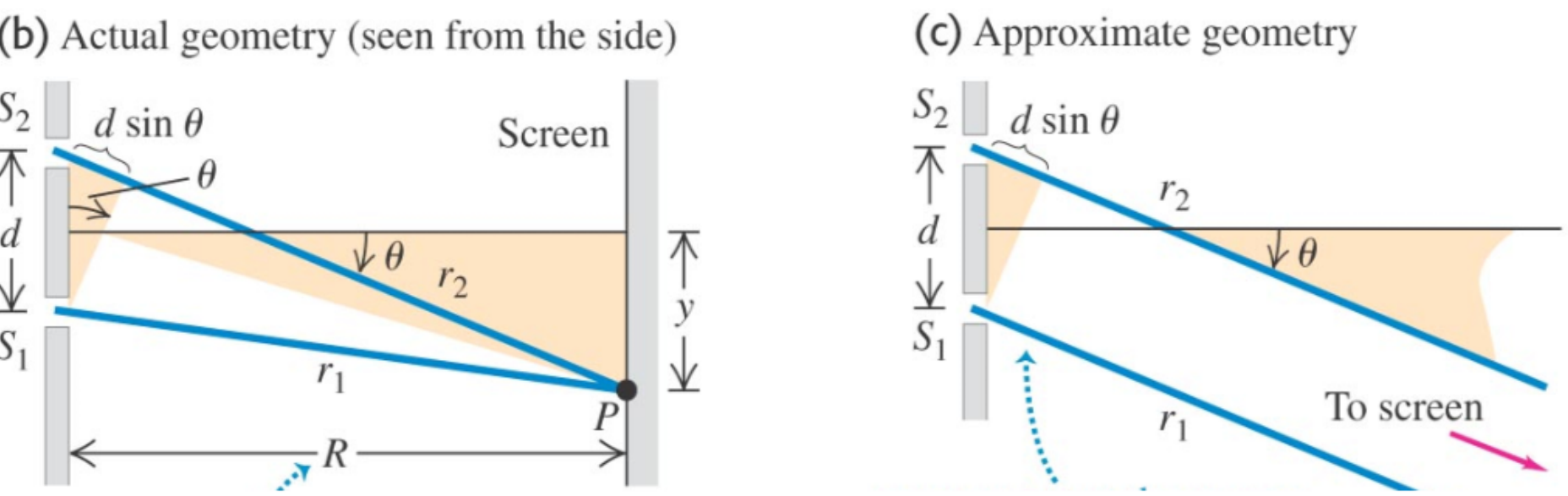
Distance between adjacent rows in array d Wavelength λ

Angle of line from surface of array to m th bright region on screen

$$\theta_a = \theta_r$$

$$2d \sin \theta = m\lambda$$

Chap 35 Interference



$$r_2 - r_1 = d \sin \theta$$

$d \sin \theta = m \lambda \quad (m = 0, \pm 1, \pm 2, \dots)$

(constructive interference, two slits)

$d \sin \theta = (m + \frac{1}{2}) \lambda \quad (m = 0, \pm 1, \pm 2, \dots)$

(destructive interference, two slits)

$y_m = R \tan \theta_m$

近似

 $y_m = R \sin \theta_m$

$y_m = R \frac{m \lambda}{d}$

(constructive interference in Young's experiment)

$E_1(t) = E \cos(\omega t + \phi)$

$E_2(t) = E \cos \omega t$

$E_P^2 = E^2 + E^2 - 2E^2 \cos(\pi - \phi)$

$= E^2 + E^2 + 2E^2 \cos \phi$

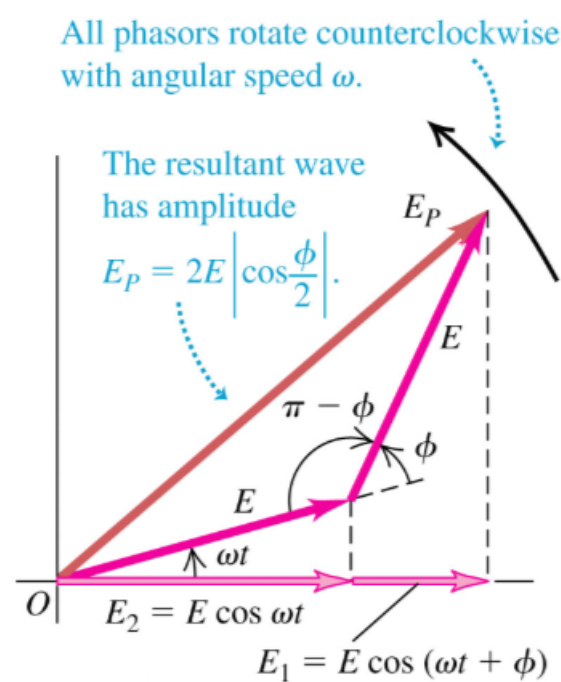
$$\text{entity } 1 + \cos \phi = 2 \cos^2(\phi/2), \text{ we obtain}$$

$$E_P^2 = 2E^2(1 + \cos \phi) = 4E^2 \cos^2\left(\frac{\phi}{2}\right)$$

$E_P = 2E \left| \cos \frac{\phi}{2} \right|$

(amplitude in two-source interference)

$$\frac{1}{2} \epsilon_0 c E^2 \text{ from each individual source.}$$



Intensity in Two-Source Interference

$$I = S_{av} = \frac{E_P^2}{2\mu_0 c} = \frac{1}{2} \sqrt{\frac{\epsilon_0}{\mu_0}} E_P^2 = \frac{1}{2} \epsilon_0 c E_P^2$$

$$I = \frac{1}{2} \epsilon_0 c E_P^2 = 2\epsilon_0 c E^2 \cos^2 \frac{\phi}{2}$$

$$\text{the maximum intensity } I_0 = 2\epsilon_0 c E^2$$

$I = I_0 \cos^2 \frac{\phi}{2}$

(intensity in two-source interference)

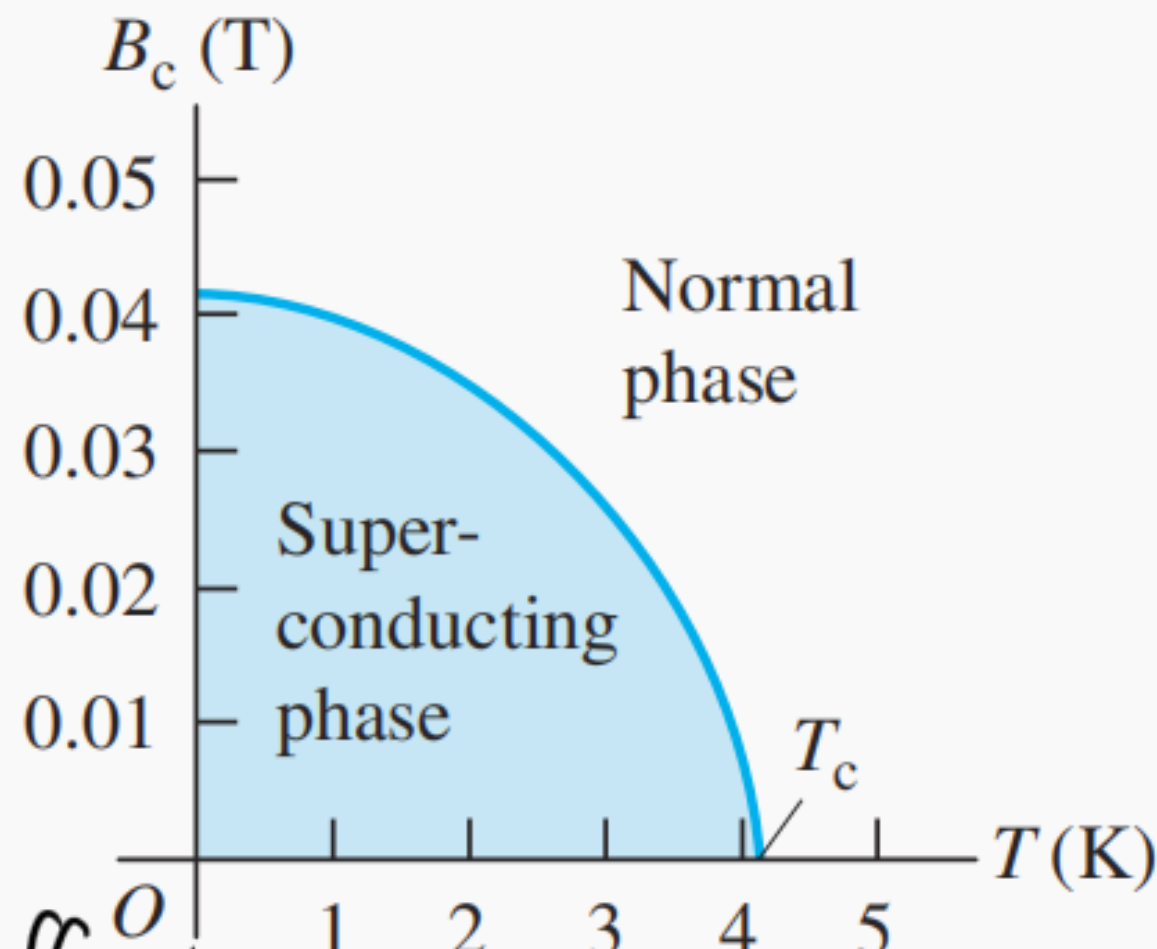
常识：① butterfly wings

② Newton's Rings

③ Interference patterns are not standing waves

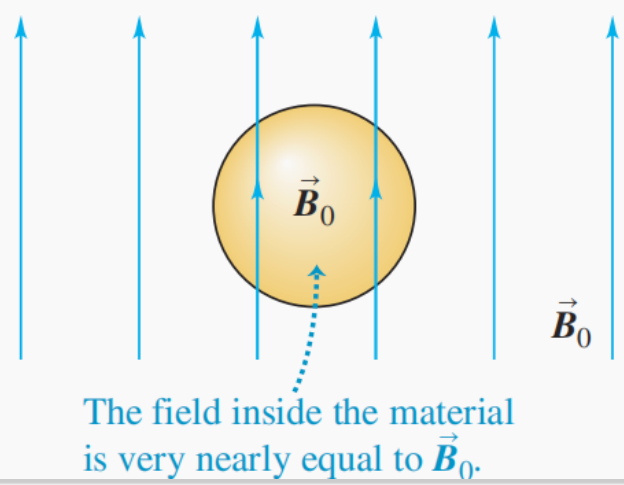
④ 超导体

Figure 29.25 Phase diagram for pure mercury, showing the critical magnetic field B_c and its dependence on temperature. Superconductivity is impossible above the critical temperature T_c . The curves for other superconducting materials are similar but with different numerical values.

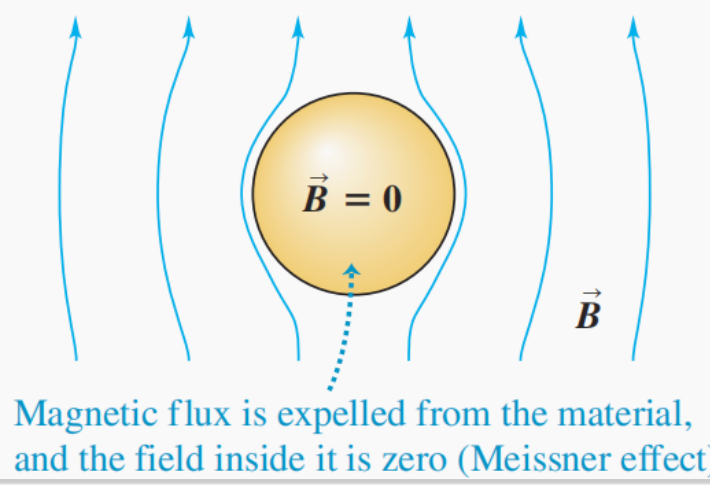


Messiner Effect

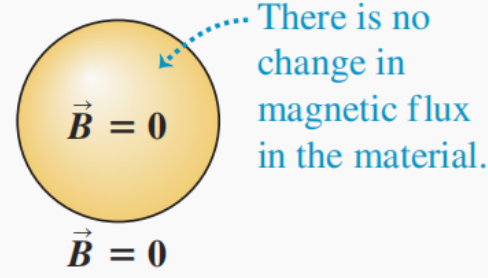
(a) Superconducting material in an external magnetic field $\vec{B}_0$ at $T > T_c$



(b) The temperature is lowered to $T < T_c$, so the material becomes superconducting.



(c) When the external field is turned off at $T < T_c$, the field is zero everywhere.



⑤ 光学补充：

如果 no atmosphere $\Rightarrow$ 地球和太空一样黑

相干光 相干时间： $T_c = \frac{1}{\Delta \nu}$
 带宽
 band width

Chap 33 & 34 Geometric Optics

1. $\frac{\sin \theta_a}{\sin \theta_b} = \frac{n_b}{n_a}$ 折射定律 Snell's Law

$n = \frac{c}{v}$ $\lambda = \frac{\lambda_0}{n}$

2. 平面 of incidence: 由入射光和界面法线确定的平面
interface 发生折射、反射的平面

"S" 偏振: E 垂直于 plane of incidence (perpendicular)

"P" 偏振: E 平行于 ... (parallel)

3. Brewster's Angle: P 偏振光完全不反射

4. Fresnel Equation:

反射系数 $r_{\perp} = \frac{E_{or}}{E_{oi}}$ $r_{\parallel} = \frac{E_{or}}{E_{oi}}$
透射系数 $t_{\perp} = \frac{E_{ot}}{E_{oi}}$ $t_{\parallel} = \frac{E_{ot}}{E_{oi}}$

$r_{\perp} + t_{\perp} = 1 \Rightarrow \frac{E_{or}}{E_{oi}} + \frac{E_{ot}}{E_{oi}} = 1$
 $\Rightarrow E_{or} + E_{ot} = E_{oi}$
 $\propto I_r + I_t = I_i$

(E 都是 field complex amplitudes)

垂直方向上 $y=0$ 处 $E_i + E_r = E_t \rightarrow$ 有极化方向的

5. total internal reflection:

$\sin \theta_{crit} = \frac{n_b}{n_a}$ ($\sin \theta_b = 1$)

6. Malus Law: $I = I_{max} \cos^2 \phi$

7. Brewster's Law: $\tan \theta_p = \frac{n_b}{n_a}$



折射率 R : 光强之比, 正比于振幅平方 $R = |r|^2$

反射率 T : $T = |t|^2$

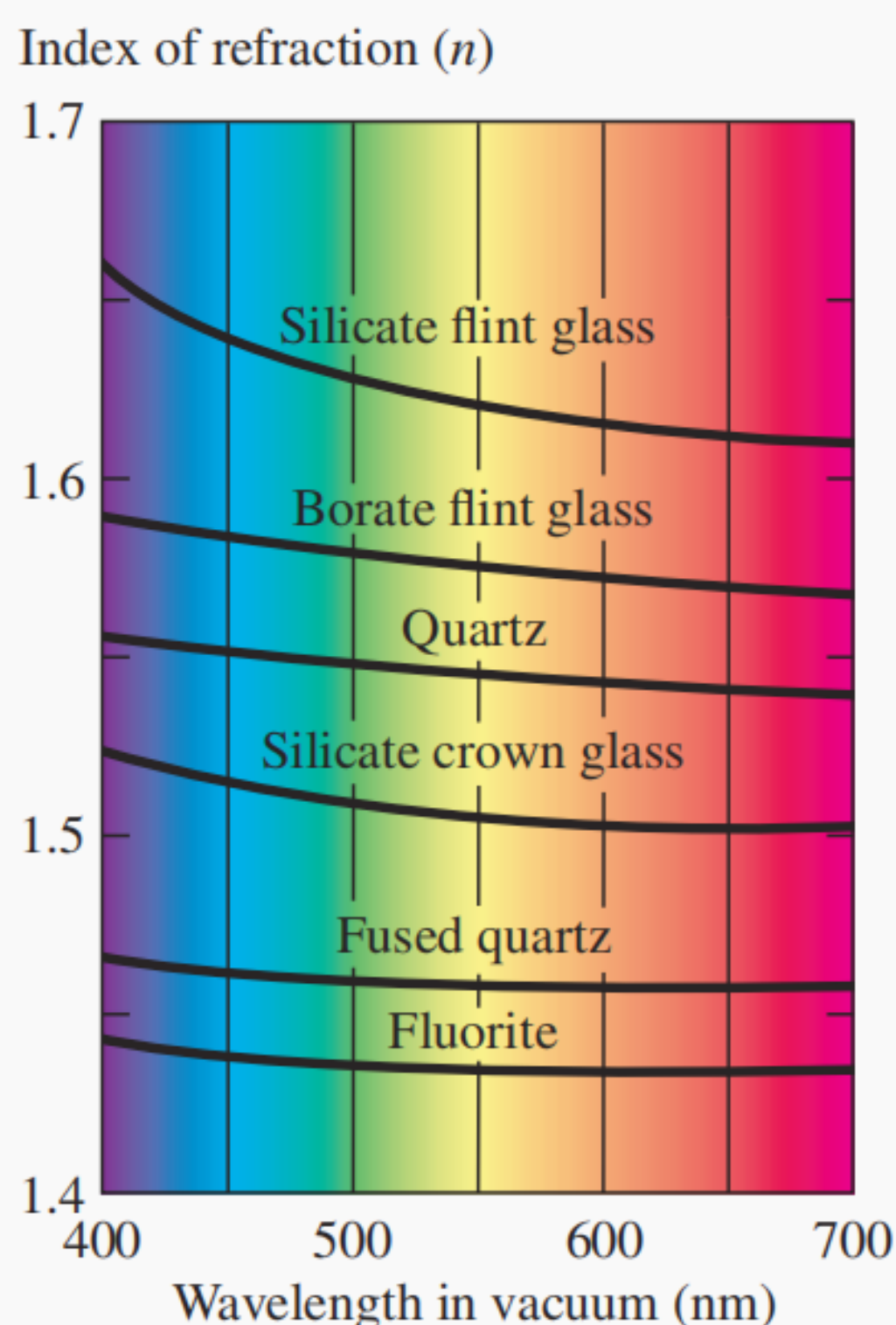
$R + T = 1$

CAUTION Wave speed and index of refraction Keep in mind that the wave speed v is inversely proportional to the index of refraction n . The larger the index of refraction in a material, the slower the wave speed in that material.

TABLE 33.1 Index of Refraction for Yellow Sodium Light, $\lambda_0 = 589$ nm

Substance	Index of Refraction
Solids	
Ice (H_2O)	1.309
Fluorite (CaF_2)	1.434
Polystyrene	1.49
Rock salt ($NaCl$)	1.544
Quartz (SiO_2)	1.544
Zircon ($ZrO_2 \cdot SiO_2$)	1.923
Diamond (C)	2.417
Fabulite ($SrTiO_3$)	2.409
Rutile (TiO_2)	2.62
Glasses (typical values)	
Crown	1.52
Light flint	1.58
Medium flint	1.62
Dense flint	1.66
Lanthanum flint	1.80
Liquids at 20°C	
Methanol (CH_3OH)	1.329
Water (H_2O)	1.333
Ethanol (C_2H_5OH)	1.36
Carbon tetrachloride (CCl_4)	1.460
Turpentine	1.472
Glycerine	1.473
Benzene	1.501
Carbon disulfide (CS_2)	1.628

Figure 33.17 Variation of index of refraction n with wavelength for different transparent materials. The horizontal axis shows the wavelength λ_0 of the light in vacuum; the wavelength in the material equal to $\lambda = \lambda_0/n$.

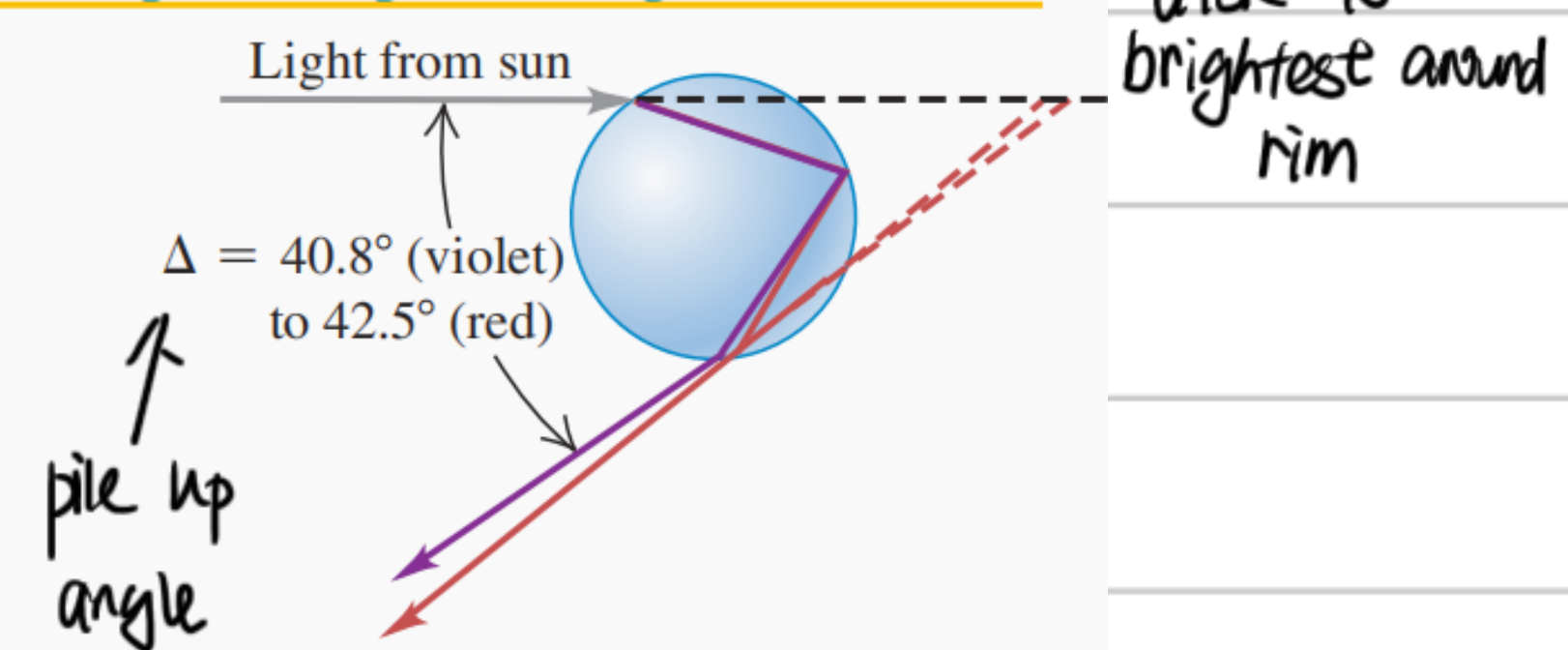


宝石: total internal reflection

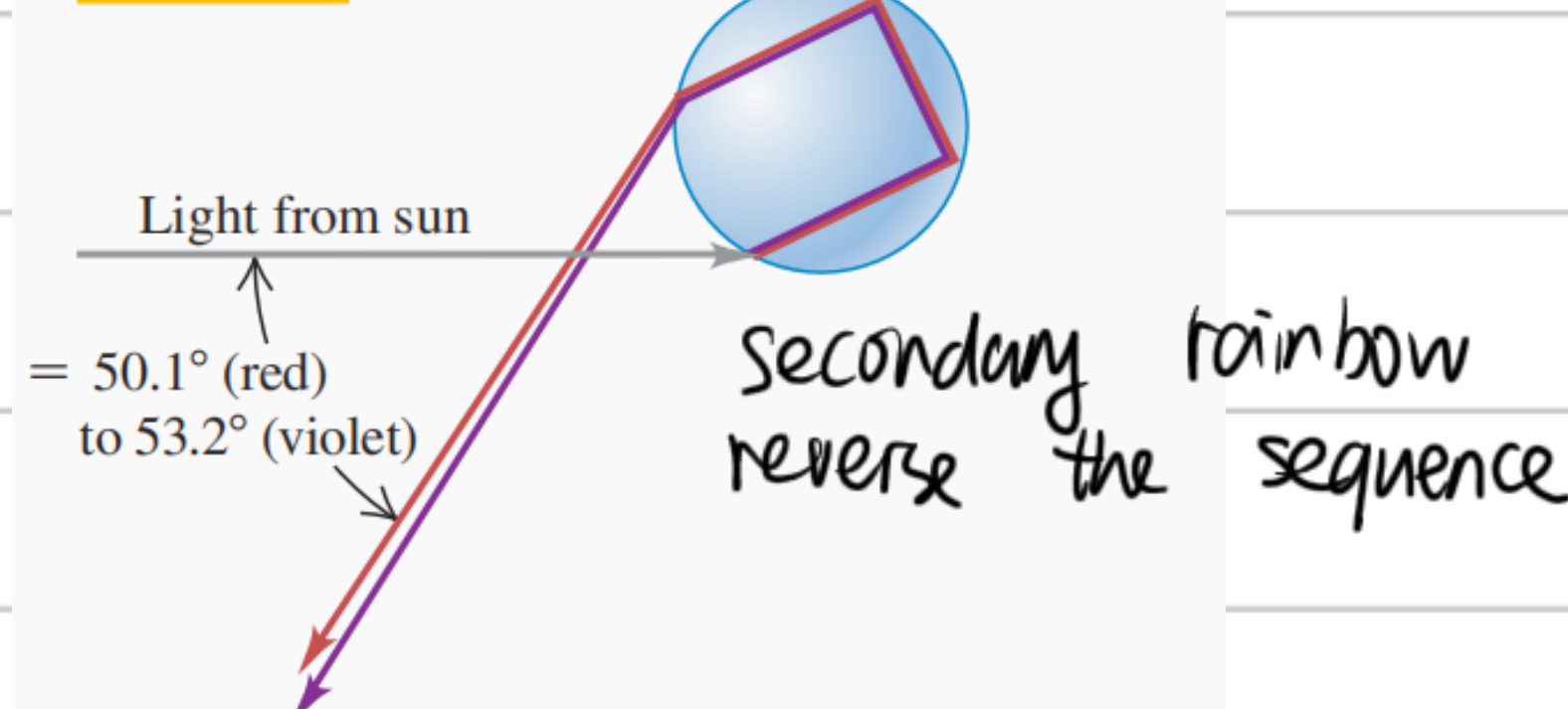
Total internal reflection also plays an important role in the design of jewelry. The brilliance of diamond is due in large measure to its very high index of refraction ($n = 2.417$) and correspondingly small critical angle. Light entering a cut diamond is

large dispersion $\Rightarrow$ multi color spectrum

(d) A primary rainbow is formed by rays that undergo two refractions and one internal reflection. The angle Δ is larger for red light than for violet.



(e) A secondary rainbow is formed by rays that undergo two refractions and two internal reflections. The angle Δ is larger for violet light than for red.

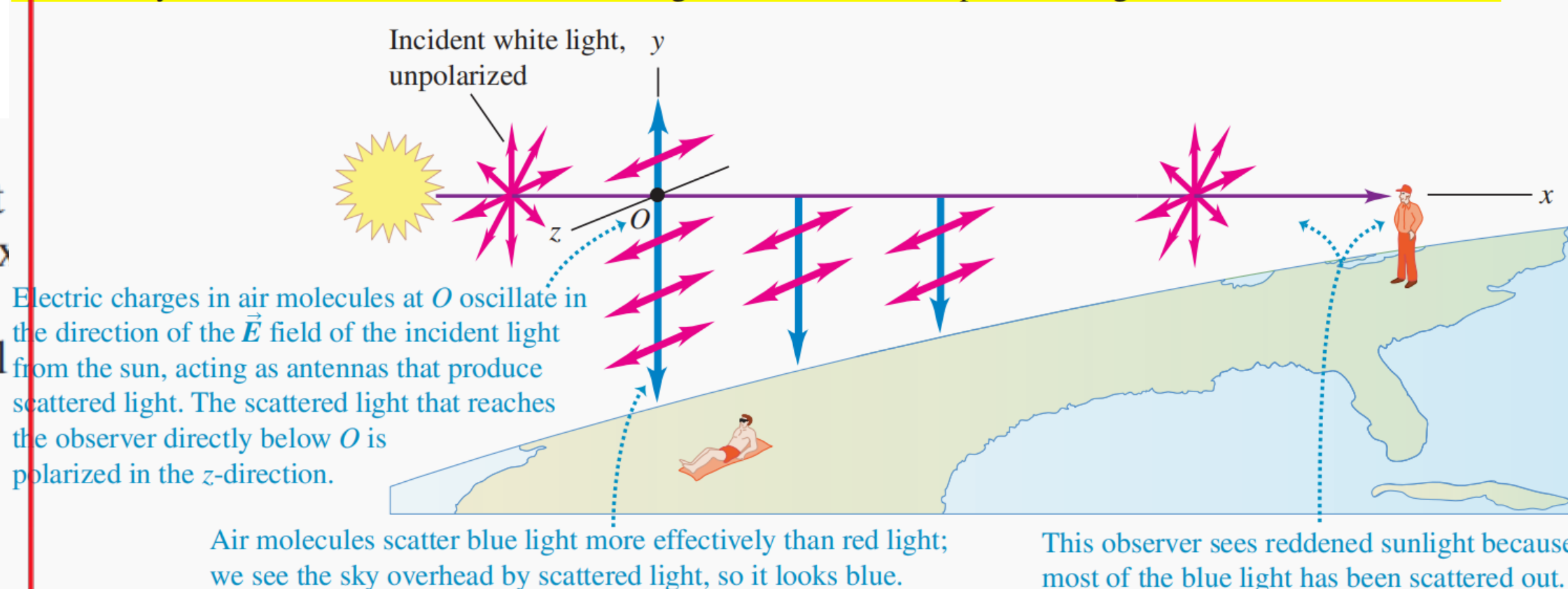


fraction, and two reflections from the back surface of the droplet (Fig. 33.19e). Each time a light ray hits the back surface, part of the light is refracted out of the drop (not shown in Fig. 33.19); after two such hits, relatively little light is left inside the drop, which is why the secondary rainbow is noticeably fainter than the primary rainbow. Just as a mirror held

沥青路面的反射极化.

Polarization by reflection is the reason polarizing filters are widely used in sunglasses (Fig. 33.25). When sunlight is reflected from a horizontal surface, the plane of incidence is vertical, and the reflected light contains a preponderance of light that is polarized in the horizontal direction. When the reflection occurs at a smooth asphalt road surface, it causes unwanted glare. To eliminate this glare, the polarizing axis of the lens material is made vertical, so very little of the horizontally polarized light reflected from the road is transmitted to the eyes. The glasses also reduce the overall intensity of the transmitted light to somewhat less than 50% of the intensity of the unpolarized incident light.

Figure 33.31 When the sunbathing observer on the left looks up, he sees blue, polarized sunlight that has been scattered by air molecules. The observer on the right sees reddened, unpolarized light when he looks at the sun.

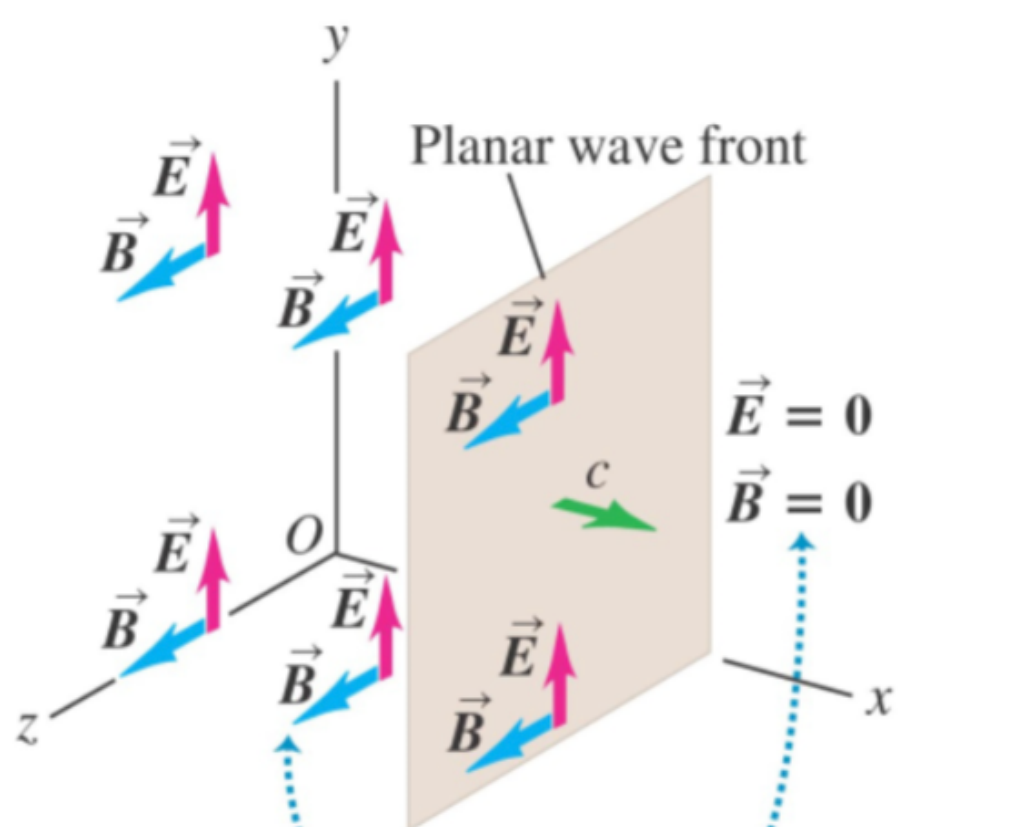


As the original beam of sunlight passes through the atmosphere, its intensity decreases as its energy goes into the scattered light. Detailed analysis of the scattering process shows that the intensity of the light scattered from air molecules increases in proportion to the fourth power of the frequency (inversely to the fourth power of the wavelength). Thus the intensity ratio for the two ends of the visible spectrum is $(750 \text{ nm}/380 \text{ nm})^4 = 15$. Roughly speaking, scattered light contains 15 times as much blue light as red, and that's why the sky is blue.

Clouds contain a high concentration of suspended water droplets or ice crystals, which also scatter light. Because of this high concentration, light passing through the cloud has many more opportunities for scattering than does light passing through a clear sky. Thus light of all wavelengths is eventually scattered out of the cloud, so the cloud looks white (Fig. 33.32). Milk looks white for the same reason; the scattering is due to fat globules suspended in the milk.

Near sunset, when sunlight has to travel a long distance through the earth's atmosphere, a substantial fraction of the blue light is removed by scattering. White light minus blue light looks yellow or red. This explains the yellow or red hue that we so often see from the setting sun (and that is seen by the observer at the far right of Fig. 33.31).

Chap 32 Electromagnetic Waves



The electric and magnetic fields are uniform behind the advancing wave front and zero in front of it.

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \quad (\text{Gauss's law for } \vec{E})$$

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (\text{Gauss's law for } \vec{B})$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_c + \epsilon_0 \frac{d\Phi_E}{dt} \right)_{\text{encl}} \quad (\text{Ampere's law})$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} \quad (\text{Faraday's law})$$

$$E = cB \quad (\text{electromagnetic wave in vacuum})$$

$$B = \epsilon_0 \mu_0 c E \quad (\text{electromagnetic wave in vacuum})$$

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \quad (\text{speed of electromagnetic waves in vacuum})$$

$$\frac{\partial^2 E_y(x, t)}{\partial x^2} = \epsilon_0 \mu_0 \frac{\partial^2 E_y(x, t)}{\partial t^2} \quad (\text{electromagnetic wave equation in vacuum})$$

$$\vec{E}(x, t) = \hat{j} E_{\text{max}} \cos(kx - \omega t)$$

$$\vec{B}(x, t) = \hat{k} B_{\text{max}} \cos(kx - \omega t)$$

$$E_{\text{max}} = c B_{\text{max}}$$

Ampere's law the displacement current

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

$$\epsilon \frac{d\Phi_E}{dt} = K \epsilon_0 \frac{d\Phi_E}{dt},$$

$$\mu = K_m \mu_0$$

$$\oint K \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl-free}}}{\epsilon_1} \quad (\text{Gauss's law in a dielectric})$$

$$\frac{\partial^2 E_y(x, t)}{\partial x^2} = \epsilon_0 \mu_0 \frac{\partial^2 E_y(x, t)}{\partial t^2} \quad (\text{electromagnetic wave equation in vacuum})$$

$$\frac{1}{v^2} = \epsilon_0 \mu_0 \quad \text{or} \quad v = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

$$v = \frac{1}{\sqrt{\epsilon \mu}} = \frac{1}{\sqrt{K K_m}} \frac{1}{\sqrt{\epsilon_0 \mu_0}} = \frac{c}{\sqrt{K K_m}} \quad (\text{speed of electromagnetic waves in a dielectric})$$

$$K_m \cong 1, \quad \frac{c}{v} = n = \sqrt{K K_m} \cong \sqrt{K}$$

Energy in electromagnetic waves

for conductor $\frac{1}{2} \epsilon_0 E^2$ 能量密度

$$\text{for inductor } \frac{1}{2} \frac{\mu_0 N^2 A}{L} B^2$$

$$\mu = \frac{B^2}{2\mu_0}$$

$$\text{for electromagnetic waves: } u = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2\mu_0} B^2 \quad B = \frac{E}{c} = \sqrt{\epsilon_0 \mu_0} E$$

Poynting Vector:

先定义-1 s : energy per unit time per unit area

$$S = \frac{1}{A} \frac{dU}{dt} = \epsilon_0 c E^2 = \frac{EB}{\mu_0}$$

$$dU = u dV = (\epsilon_0 E^2) (A c dt)$$

定义 Poynting Vector:

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

$$P = \oint \vec{S} \cdot d\vec{A}$$

pointing to the direction of propagation

$$I = S_{\text{av}} = \frac{E_{\text{max}} B_{\text{max}}}{2\mu_0} = \frac{E_{\text{max}}^2}{2\mu_0 c} = \frac{1}{2} \sqrt{\frac{\epsilon_0}{\mu_0}} E_{\text{max}}^2 = \frac{1}{2} \epsilon_0 c E_{\text{max}}^2$$

electromagnetic momentum

$$\frac{1}{A} \frac{dp}{dt} = \frac{S}{c} = \frac{EB}{\mu_0 c}$$

pressure

change 对于不同材料上是不一样的

mirror

black

$$P_1 = 2P_2$$

驻波条件: 已在反射导体面处为0; 距离半波长整数倍

Chap 29 Electromagnetic Induction

$$1. \text{ Induced EMF } \mathcal{E} = \int \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

感应电流正方向: 右手四指方向, 但是负值

$$\mathcal{E} = -\frac{d\Phi}{dt} = -\frac{L \Delta I}{\Delta t} = -BLV$$

$$d\vec{p} = I d\vec{l} \times \vec{B}$$

$$I = \frac{|\mathcal{E}|}{R} = \frac{BLV}{R}$$

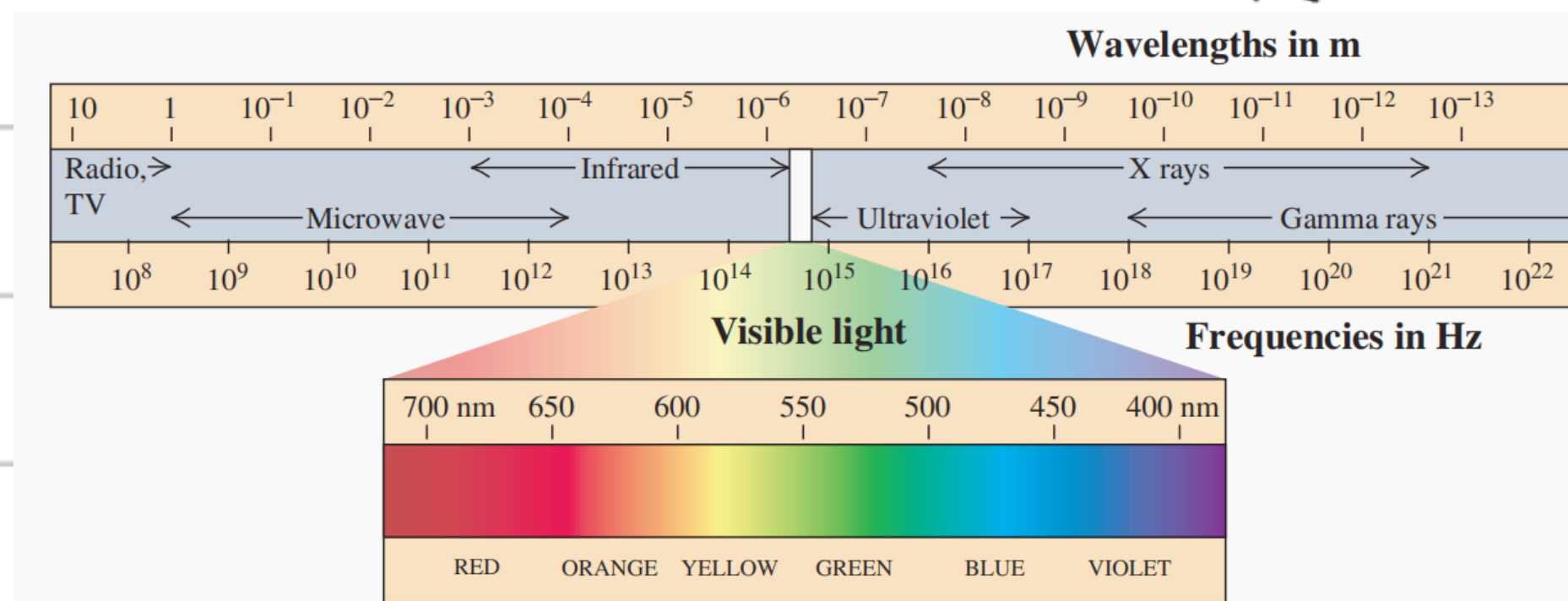
耗散功率:

$$P_{\text{dissipated}} = I^2 R = \left(\frac{BLV}{R} \right)^2 R = \frac{B^2 L^2 V^2}{R}$$

做功功率

$$F = ILB = \frac{BLV}{R} LB = \frac{B^2 L^2 V}{R}$$

$$P_{\text{applied}} = FV = \frac{B^2 L^2 V^2}{R}$$



Eddy's Current: 安培